

Project title

Design and Analysis of a Low-Power Conditional-Capture Flip-Flop in 32 nm CMOS

Project description

In this project, each group will design, implement, and evaluate a low-power conditional-capture flip-flop (CCFF) in a 32 nm CMOS technology, targeting at least 20–30% clock power reduction compared with a conventional master–slave flip-flop under similar operating conditions. The key idea of the CCFF is to use data-dependent control logic that prevents unnecessary internal switching when the input data does not change ($D = Q$), thereby statistically reducing dynamic power while maintaining acceptable timing (setup/hold, clk-to-Q) and reasonable area overhead.

Students will:

1. Study classical and modern low-power flip-flop structures, including conditional-capture, conditional-precharge, conditional-discharge, and pulse-triggered designs.
2. Propose a CCFF architecture with simplified logic suitable for implementation using Electric (schematic and layout) and transistor- or gate-level simulation.
3. Perform pre-layout and post-layout simulations (timing and power) on both the proposed CCFF and a reference standard flip-flop, and build a small register bank (e.g., 16–32 bits) to evaluate clock power, total dynamic power, and area under different activity patterns.
4. Compare your results against these three published designs:
 - Reference 1: K. Kang et al., "Conditional-Capture Flip-Flop for Statistical Power Reduction," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 16, no. 10, pp. 1335–1344, Oct. 2008.
(Available: <https://ieeexplore.ieee.org/document/4626154>)[3]
 - Reference 2: A. Mahmood et al., "High-Performance and Low-Power Conditional Discharge Flip-Flop," *IEEE Transactions on VLSI Systems*, vol. 12, no. 5, May 2004.
(Available: <https://ieeexplore.ieee.org/document/1317825>)[1]
 - Reference 3: S. Roy and S. C. Prasad, "Low-Power Pulse Triggered Flip-Flop Design with Conditional Discharge Technique," *International Journal of VLSI Design & Communication Systems*, vol. 5, no. 4, Aug. 2014. (Available at research databases)

Create a comparison table in your report showing: clock power, total dynamic power, area, setup time, hold time, and clock-to-Q delay for your design and the three reference designs at typical corners.

Deliverables

Presentation/Discussion (PPT) – 50 points

Design (5 points):

- Clear explanation of CCFF architecture, control logic, and data-dependency mechanism.
- Comparison with standard flip-flop structure.

Simulation (Pre-Layout) (5 points):

- AC and transient analysis results showing timing margins.
- Power analysis (clock power, dynamic power per switching activity).

Layout (5 points):

- Stick diagram or layout topology showing area optimization.
- Device sizing and placement strategy.

Simulation (Post-Layout) (5 points):

- Timing and power results after parasitic extraction.
- Comparison of pre- vs post-layout performance degradation.

Innovation & Results (Area, Power, Speed) (10 points × 3 = 30 points total, distributed as follows):

- Area Innovation (10 points): Demonstrate area reduction or comparable area vs. standard FF and reference designs.
- Power Innovation (10 points): Show $\geq 20\text{--}30\%$ clock power saving and quantify total energy per operation.
- Speed Innovation (10 points): Maintain or improve timing (setup/hold/clk-to-Q) compared to reference designs.

Report (IEEE TCAS-II Format) – 50 points

Maximum length: 5 pages (IEEE two-column format)

Template: Use the official IEEE TCAS-II template available at:

- IEEE TCAS-II Author Guidelines: <https://ieee-cas.org/publication/TCAS-II/guidelines-author>
- IEEE TCAS-II Manuscript Submission Guide: <https://ieee-cas.org/publication/TCAS-II/tcas-ii-manuscript-submission-guide>
- LaTeX/Word Templates (IEEE): <https://www.overleaf.com/gallery/tagged/ieee-official>
- Authorea TCAS-II Template: <https://www.authorea.com/templates/ieee-transactions-on-circuits-and-systems-ii-express-briefs>

Design Section (10 points):

- Motivation, problem statement, and proposed CCFF architecture clearly described.
- Block diagram, gate-level schematic, and operating principle.
- Comparison to standard master–slave and reference CCFF designs.

Simulation (Pre-Layout) Section (10 points):

- Transistor-level netlist and simulation setup (supply voltage, process corner, temperature).
- Timing analysis (setup time, hold time, clk-to-Q), power analysis, and waveforms.
- Results presented in table and graph format.

Layout Section (10 points):

- Physical layout in Electric, parasitic extraction methodology.
- Layout area breakdown (gates, interconnects).
- Design rules and technology node constraints addressed.

Simulation (Post-Layout) Section (10 points):

- Post-layout timing and power results (with parasitic capacitances).
- Comparison of pre- vs post-layout performance.
- Analysis of performance degradation and margin assessment.

Innovation, Results & Comparison (10 points):

- Clear comparison table showing your design vs. Reference 1, Reference 2, and Reference 3 across: clock power, total dynamic power, area, setup time, hold time, and clk-to-Q delay.
- From calculation (e.g., energy per operation, power×area product).
- Discussion of advantages and limitations.
- Conclusion and future work.

Submission deadline

Last day of classes – no extensions.

Submit both:

1. PPT presentation file
2. Full paper (PDF) in IEEE TCAS-II format

Papers not following the TCAS-II template or missing comparison to the three specified references will have points deducted.

Key reminders for students:

- Use Electric for schematic and layout design.
- Export netlists to LTspice (or equivalent 32 nm-compatible simulator) for pre-layout simulations.
- Run post-layout simulations with extracted parasitics to validate your design.
- Build a small test structure (16–32 bit register bank) to measure realistic power savings.
- Follow IEEE TCAS-II style, citation format, and figure/table numbering standards.